

# 1D CNN — Architecture Variant Comparison

Generated 2026-05-08 15:11:34

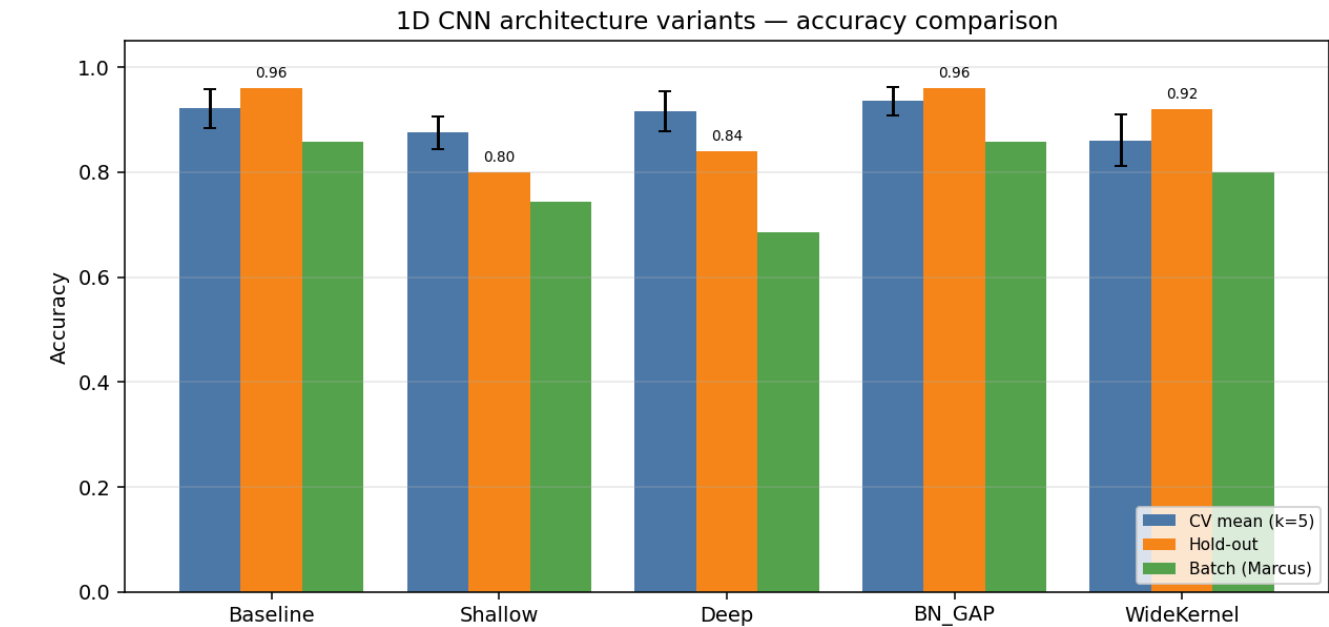
## Data & configuration

|                     |                                                                                                             |
|---------------------|-------------------------------------------------------------------------------------------------------------|
| Training path 1     | /home/jestin/ThesisRepo/ML/NewTestData/Alan/Dynamic                                                         |
| Training path 2     | /home/jestin/ThesisRepo/ML/NewTestData/Anghad/Dynamic                                                       |
| Training path 3     | /home/jestin/ThesisRepo/ML/NewTestData/Harry/Dynamic                                                        |
| Training path 4     | /home/jestin/ThesisRepo/ML/NewTestData/Bella/Dynamic                                                        |
| Training path 5     | /home/jestin/ThesisRepo/ML/NewTestData/Jestin/Dynamic                                                       |
| Training path 6     | /home/jestin/ThesisRepo/ML/NewTestData/Joselyn/Dynamic                                                      |
| Training path 7     | /home/jestin/ThesisRepo/ML/NewTestData/Raquel/Dynamic                                                       |
| Training path 8     | /home/jestin/ThesisRepo/ML/NewTestData/Tash/Dynamic                                                         |
| Batch-test root     | /home/jestin/ThesisRepo/ML/NewTestData/Marcus/Dynamic                                                       |
| Total trials        | 241 (train pool 216, hold-out 25)                                                                           |
| Classes             | 7: Double_Lower, Double_Nothing, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll |
| Sensor channels     | 176 (sequence length 90)                                                                                    |
| Resample / filter   | 90 steps · Butterworth cutoff 10.0 Hz, order 4                                                              |
| Normalisation       | minmax                                                                                                      |
| CV folds · shuffle  | 5 · True                                                                                                    |
| Augmentation        | on · copies=5 · amplitude_scale, baseline_offset                                                            |
| Epochs · batch size | 15 · 16                                                                                                     |

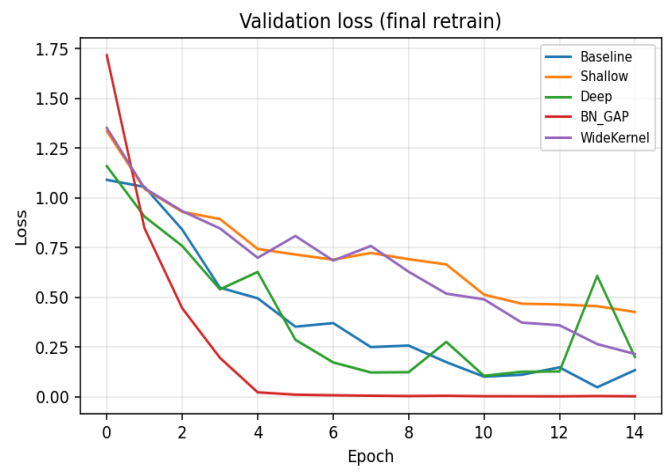
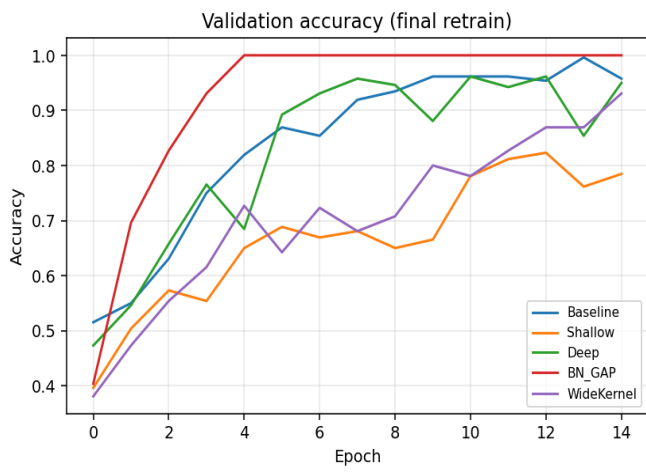
## Variant comparison (summary)

| Variant    | Params  | CV mean (k=5) | CV std | Hold-out acc | Hold-out loss | Batch acc | Batch n |
|------------|---------|---------------|--------|--------------|---------------|-----------|---------|
| Baseline   | 113,255 | 0.9214        | 0.0376 | 0.9600       | 0.1717        | 0.8571    | 30/35   |
| Shallow    | 72,487  | 0.8752        | 0.0307 | 0.8000       | 0.4891        | 0.7429    | 26/35   |
| Deep       | 204,007 | 0.9166        | 0.0379 | 0.8400       | 0.7520        | 0.6857    | 24/35   |
| BN_GAP     | 57,319  | 0.9353        | 0.0267 | 0.9600       | 0.1718        | 0.8571    | 30/35   |
| WideKernel | 131,687 | 0.8609        | 0.0492 | 0.9200       | 0.2524        | 0.8000    | 28/35   |

Best on hold-out: **Baseline** · Best on CV: **BN\_GAP**



## Training curves (final retrain on full training pool)



# Variant: Baseline

Conv1D(32,k=4)→Pool→Conv1D(64,k=4)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

|                          |                 |
|--------------------------|-----------------|
| Trainable params         | 113,255         |
| CV mean ± std            | 0.9214 ± 0.0376 |
| Hold-out test acc / loss | 0.9600 · 0.1717 |
| Batch-test acc           | 0.8571 (30/35)  |

## Cross-validation folds

| Fold | Train (post-aug) | Val | Val acc | Best val | Final val |
|------|------------------|-----|---------|----------|-----------|
| 1    | 1032             | 44  | 0.9091  | 0.9318   | 0.9091    |
| 2    | 1038             | 43  | 0.9535  | 0.9535   | 0.9535    |
| 3    | 1038             | 43  | 0.9767  | 0.9767   | 0.9767    |
| 4    | 1038             | 43  | 0.8837  | 0.9070   | 0.8837    |
| 5    | 1038             | 43  | 0.8837  | 0.9070   | 0.8837    |

## Per-class hold-out metrics

| Class                | Precision | Recall | F1    | Support |
|----------------------|-----------|--------|-------|---------|
| Double_Lower         | 0.800     | 1.000  | 0.889 | 4       |
| Double_Nothing       | 1.000     | 0.667  | 0.800 | 3       |
| Double_Pistol_Recoil | 1.000     | 1.000  | 1.000 | 4       |
| Double_Raise         | 1.000     | 1.000  | 1.000 | 4       |
| Double_Wiggle        | 1.000     | 1.000  | 1.000 | 3       |
| Double_cmere         | 1.000     | 1.000  | 1.000 | 4       |
| Drum_Roll            | 1.000     | 1.000  | 1.000 | 3       |
| macro avg            | 0.971     | 0.952  | 0.956 | 25      |
| weighted avg         | 0.968     | 0.960  | 0.958 | 25      |

## Batch test per class (Marcus)

| Class                | Correct | Total | Accuracy |
|----------------------|---------|-------|----------|
| Double_Lower         | 5       | 5     | 1.000    |
| Double_Nothing       | 0       | 5     | 0.000    |
| Double_Pistol_Recoil | 5       | 5     | 1.000    |
| Double_Raise         | 5       | 5     | 1.000    |
| Double_Wiggle        | 5       | 5     | 1.000    |
| Double_cmere         | 5       | 5     | 1.000    |
| Drum_Roll            | 5       | 5     | 1.000    |
| Overall              | 30      | 35    | 0.857    |

# Variant: Shallow

Single Conv block: Conv1D(32,k=5)→Pool→Flatten→Dense(32)→Dropout(0.3)→Softmax

|                          |                 |
|--------------------------|-----------------|
| Trainable params         | 72,487          |
| CV mean ± std            | 0.8752 ± 0.0307 |
| Hold-out test acc / loss | 0.8000 · 0.4891 |
| Batch-test acc           | 0.7429 (26/35)  |

## Cross-validation folds

| Fold | Train (post-aug) | Val | Val acc | Best val | Final val |
|------|------------------|-----|---------|----------|-----------|
| 1    | 1032             | 44  | 0.8409  | 0.8864   | 0.8409    |
| 2    | 1038             | 43  | 0.9070  | 0.9535   | 0.9070    |
| 3    | 1038             | 43  | 0.9070  | 0.9070   | 0.9070    |
| 4    | 1038             | 43  | 0.8372  | 0.8372   | 0.8372    |
| 5    | 1038             | 43  | 0.8837  | 0.8837   | 0.8837    |

## Per-class hold-out metrics

| Class                | Precision | Recall | F1    | Support |
|----------------------|-----------|--------|-------|---------|
| Double_Lower         | 1.000     | 0.750  | 0.857 | 4       |
| Double_Nothing       | 1.000     | 0.333  | 0.500 | 3       |
| Double_Pistol_Recoil | 0.800     | 1.000  | 0.889 | 4       |
| Double_Raise         | 1.000     | 1.000  | 1.000 | 4       |
| Double_Wiggle        | 0.429     | 1.000  | 0.600 | 3       |
| Double_cmere         | 1.000     | 1.000  | 1.000 | 4       |
| Drum_Roll            | 1.000     | 0.333  | 0.500 | 3       |
| macro avg            | 0.890     | 0.774  | 0.764 | 25      |
| weighted avg         | 0.899     | 0.800  | 0.791 | 25      |

## Batch test per class (Marcus)

| Class                | Correct | Total | Accuracy |
|----------------------|---------|-------|----------|
| Double_Lower         | 5       | 5     | 1.000    |
| Double_Nothing       | 2       | 5     | 0.400    |
| Double_Pistol_Recoil | 5       | 5     | 1.000    |
| Double_Raise         | 5       | 5     | 1.000    |
| Double_Wiggle        | 4       | 5     | 0.800    |
| Double_cmere         | 5       | 5     | 1.000    |
| Drum_Roll            | 0       | 5     | 0.000    |
| Overall              | 26      | 35    | 0.743    |

# Variant: Deep

Three Conv blocks: Conv1D(32,4)→Conv1D(64,4)→Conv1D(128,3)→Flatten→Dense(128)→Dropout(0.4)→Softmax

|                          |                 |
|--------------------------|-----------------|
| Trainable params         | 204,007         |
| CV mean ± std            | 0.9166 ± 0.0379 |
| Hold-out test acc / loss | 0.8400 · 0.7520 |
| Batch-test acc           | 0.6857 (24/35)  |

## Cross-validation folds

| Fold | Train (post-aug) | Val | Val acc | Best val | Final val |
|------|------------------|-----|---------|----------|-----------|
| 1    | 1032             | 44  | 0.9318  | 0.9773   | 0.9318    |
| 2    | 1038             | 43  | 0.9535  | 0.9767   | 0.9535    |
| 3    | 1038             | 43  | 0.9535  | 0.9535   | 0.9535    |
| 4    | 1038             | 43  | 0.8605  | 0.9302   | 0.8605    |
| 5    | 1038             | 43  | 0.8837  | 0.8837   | 0.8837    |

## Per-class hold-out metrics

| Class                | Precision | Recall | F1    | Support |
|----------------------|-----------|--------|-------|---------|
| Double_Lower         | 1.000     | 1.000  | 1.000 | 4       |
| Double_Nothing       | 0.500     | 0.667  | 0.571 | 3       |
| Double_Pistol_Recoil | 0.667     | 1.000  | 0.800 | 4       |
| Double_Raise         | 1.000     | 1.000  | 1.000 | 4       |
| Double_Wiggle        | 1.000     | 0.667  | 0.800 | 3       |
| Double_cmere         | 1.000     | 1.000  | 1.000 | 4       |
| Drum_Roll            | 1.000     | 0.333  | 0.500 | 3       |
| macro avg            | 0.881     | 0.810  | 0.810 | 25      |
| weighted avg         | 0.887     | 0.840  | 0.833 | 25      |

## Batch test per class (Marcus)

| Class                | Correct | Total | Accuracy |
|----------------------|---------|-------|----------|
| Double_Lower         | 5       | 5     | 1.000    |
| Double_Nothing       | 1       | 5     | 0.200    |
| Double_Pistol_Recoil | 5       | 5     | 1.000    |
| Double_Raise         | 5       | 5     | 1.000    |
| Double_Wiggle        | 3       | 5     | 0.600    |
| Double_cmere         | 5       | 5     | 1.000    |
| Drum_Roll            | 0       | 5     | 0.000    |
| Overall              | 24      | 35    | 0.686    |

# Variant: BN\_GAP

Conv+BN+ReLU ×3 with GlobalAveragePooling1D head — fewer params, less overfit

|                          |                 |
|--------------------------|-----------------|
| Trainable params         | 57,319          |
| CV mean ± std            | 0.9353 ± 0.0267 |
| Hold-out test acc / loss | 0.9600 · 0.1718 |
| Batch-test acc           | 0.8571 (30/35)  |

## Cross-validation folds

| Fold | Train (post-aug) | Val | Val acc | Best val | Final val |
|------|------------------|-----|---------|----------|-----------|
| 1    | 1032             | 44  | 0.9091  | 0.9773   | 0.9091    |
| 2    | 1038             | 43  | 0.9767  | 0.9767   | 0.9767    |
| 3    | 1038             | 43  | 0.9535  | 0.9767   | 0.9535    |
| 4    | 1038             | 43  | 0.9070  | 0.9302   | 0.9070    |
| 5    | 1038             | 43  | 0.9302  | 0.9767   | 0.9302    |

## Per-class hold-out metrics

| Class                | Precision | Recall | F1    | Support |
|----------------------|-----------|--------|-------|---------|
| Double_Lower         | 1.000     | 1.000  | 1.000 | 4       |
| Double_Nothing       | 1.000     | 0.667  | 0.800 | 3       |
| Double_Pistol_Recoil | 0.800     | 1.000  | 0.889 | 4       |
| Double_Raise         | 1.000     | 1.000  | 1.000 | 4       |
| Double_Wiggle        | 1.000     | 1.000  | 1.000 | 3       |
| Double_cmere         | 1.000     | 1.000  | 1.000 | 4       |
| Drum_Roll            | 1.000     | 1.000  | 1.000 | 3       |
| macro avg            | 0.971     | 0.952  | 0.956 | 25      |
| weighted avg         | 0.968     | 0.960  | 0.958 | 25      |

## Batch test per class (Marcus)

| Class                | Correct | Total | Accuracy |
|----------------------|---------|-------|----------|
| Double_Lower         | 5       | 5     | 1.000    |
| Double_Nothing       | 5       | 5     | 1.000    |
| Double_Pistol_Recoil | 5       | 5     | 1.000    |
| Double_Raise         | 5       | 5     | 1.000    |
| Double_Wiggle        | 5       | 5     | 1.000    |
| Double_cmere         | 5       | 5     | 1.000    |
| Drum_Roll            | 0       | 5     | 0.000    |
| Overall              | 30      | 35    | 0.857    |

# Variant: WideKernel

Conv1D(32,k=8)→Pool→Conv1D(64,k=8)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

|                          |                 |
|--------------------------|-----------------|
| Trainable params         | 131,687         |
| CV mean ± std            | 0.8609 ± 0.0492 |
| Hold-out test acc / loss | 0.9200 · 0.2524 |
| Batch-test acc           | 0.8000 (28/35)  |

## Cross-validation folds

| Fold | Train (post-aug) | Val | Val acc | Best val | Final val |
|------|------------------|-----|---------|----------|-----------|
| 1    | 1032             | 44  | 0.9091  | 0.9773   | 0.9091    |
| 2    | 1038             | 43  | 0.8140  | 0.9535   | 0.8140    |
| 3    | 1038             | 43  | 0.9070  | 0.9535   | 0.9070    |
| 4    | 1038             | 43  | 0.8837  | 0.8837   | 0.8837    |
| 5    | 1038             | 43  | 0.7907  | 0.9070   | 0.7907    |

## Per-class hold-out metrics

| Class                | Precision | Recall | F1    | Support |
|----------------------|-----------|--------|-------|---------|
| Double_Lower         | 0.800     | 1.000  | 0.889 | 4       |
| Double_Nothing       | 1.000     | 0.333  | 0.500 | 3       |
| Double_Pistol_Recoil | 1.000     | 1.000  | 1.000 | 4       |
| Double_Raise         | 1.000     | 1.000  | 1.000 | 4       |
| Double_Wiggle        | 1.000     | 1.000  | 1.000 | 3       |
| Double_cmere         | 0.800     | 1.000  | 0.889 | 4       |
| Drum_Roll            | 1.000     | 1.000  | 1.000 | 3       |
| macro avg            | 0.943     | 0.905  | 0.897 | 25      |
| weighted avg         | 0.936     | 0.920  | 0.904 | 25      |

## Batch test per class (Marcus)

| Class                | Correct | Total | Accuracy |
|----------------------|---------|-------|----------|
| Double_Lower         | 5       | 5     | 1.000    |
| Double_Nothing       | 1       | 5     | 0.200    |
| Double_Pistol_Recoil | 5       | 5     | 1.000    |
| Double_Raise         | 5       | 5     | 1.000    |
| Double_Wiggle        | 5       | 5     | 1.000    |
| Double_cmere         | 5       | 5     | 1.000    |
| Drum_Roll            | 2       | 5     | 0.400    |
| Overall              | 28      | 35    | 0.800    |